

Name of the Experiment : Implementation of S-R Flipflop circuit.

Object : The object of this experiment is to implement a S-R flip-flop circuit.

Theory : A S-R flip-flop is a sequential circuit consists of three inputs set (S), reset and clock-pulse and two output Q and $\bar{Q}$. It is constructed by using a NAND latch, two NAND gates and clock-pulse. It is triggered by positive-going-edge of the clock signal in NAND latch, two NAND gates cross-coupled, so that the output of NAND-1 is connected to one of the inputs of NAND-2 and vice versa. S and clk are inputs of NAND and the output of this is one of the inputs of NAND-1 of the latch. Similarly, $\bar{S}$ and clk are the input

of one NAND gate and the output of this gate is one of the inputs of NAND-2 of the latch.

Then the outputs of NAND-1 and NAND-2 are the flip-flop's output Q and $\bar{Q}$.

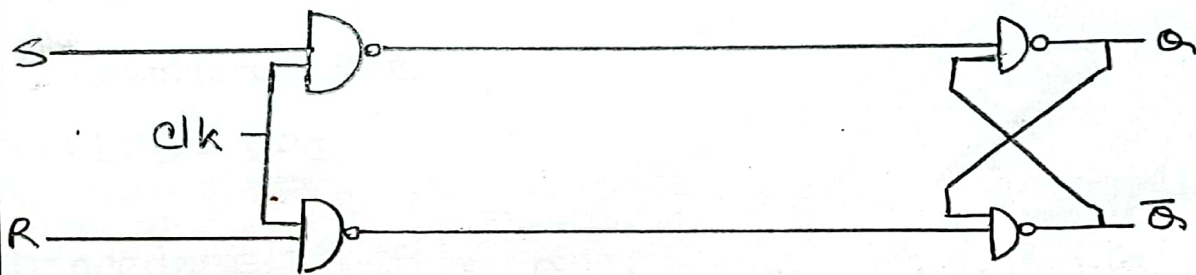


Fig : S-R flip-flop

Truth table :

Input			Output		Comment
clk	S	R	Q	$\bar{Q}$	
↑	0	0	Q	$\bar{Q}$	No change
↑	0	1	0	1	Reset
↑	1	0	1	0	set
↑	1	1	?	?	Invalid

Apparatus :

- i) NAND-gate .
- ii) Clock - pulse .
- iii) Bread-Board .
- iv) Connecting wire .
- v) Resistor - 2PC .
- vi) LED - 2PC .

Procedure :

At first, we have to identify the pin numbers of the IC and then set it on the bread-board. Connect the wire and clock pulse in the inputs pin and make the pulse circuit on bread board. Output pin connected to the resistor and then to the LED. All the Vee and GND connected with the power supply. According to the truth table, we give the inputs and then check the outputs.

Result :

CLK	S	R	Q	LED	Q	LED
↑	0	0				
↑	0	1	0	OFF	1	ON
↑	1	0	1	ON	0	OFF
↑	1	1				

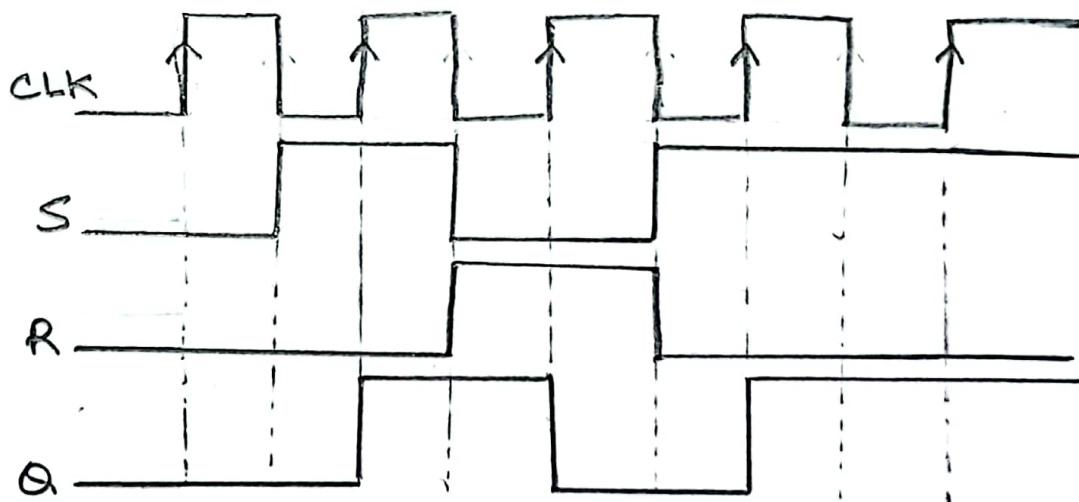


Fig: Clock Diagram of S-R flip-flop.

Discussion :

In this experiment, we have checked all the gates of IC's and found out the active gates. We also found out the position at which the IC's have proper supply. We connected the Vee and GND properly. We didn't touch any wire where it was open.

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